

PHYSICS

Q1 A particle is projected with speed v_0 at an angle of 45° from horizontal. Find angular momentum of the particle at highest point of trajectory.

Sol. $\vec{L} = m(\vec{r} \times \vec{V})$

$$= mv\gamma_{\perp}$$

$$= \frac{m v_0}{\sqrt{2}} \times \frac{v_0^2}{4g}$$

$$= \frac{m v_0^3}{4\sqrt{2}g}$$

Q2 A wire of length l having total resistance R is bent to form an equilateral triangle, Find equivalent resistance between any two vertices of triangle.

Sol. $R = \frac{\rho \ell}{A}$

$$R \propto l$$

$$R_{eq} = \frac{6 \times 3}{6 + 3} = 2\Omega$$

Q3 Given current $I = I_A \sin \omega t + I_B \cos \omega t$. Find the rms value of current.

Sol. $I = I_A \sin \omega t + I_B \cos \omega t$

$$\sqrt{I_A^2 + I_B^2} \sin(\omega t + \phi)$$

$$I_{rms} = \sqrt{\frac{I_A^2 + I_B^2}{2}}$$

Q4 Given $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{B} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{A} \cdot \vec{C} = 5$, $\vec{B} \perp \vec{C}$ and $\vec{A}, \vec{B}, \vec{C}$ are coplanar then find $\vec{C}$.

Sol. $\vec{C} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{A} \cdot \vec{C} = 5 = x + 2y + 3 \times 3 = 5$$

$$\vec{B} \cdot \vec{C} = 0 \Rightarrow 3x + y - 3 = 0$$

$$(\vec{A} \times \vec{B}) \cdot \vec{C} = 0$$

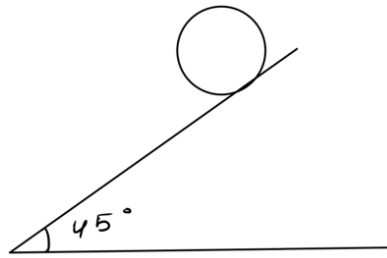
$$(-5\hat{i} + 10\hat{j} - 5\hat{k}) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) = 0$$

$$-5x + 10y - 5z = 0$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & 1 & -1 \end{vmatrix}$$

$$= \lambda[-5] + \hat{i}[10] + \hat{k}[-5]$$

- Q5** Find out the linear acceleration of a solid cylinder of mass m and radius R rolling down an inclined plane of inclination 45°



Sol.

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$$

$$a = \frac{g \sin 45^\circ}{1 + \frac{2mR^2}{5mR^2}}$$

$$= \frac{5}{7} \times 10 \times \frac{1}{\sqrt{2}}$$

$$= \frac{25}{7} \sqrt{2} \text{ m / sec}^2$$

- Q6** A satellite revolving in a circular orbit of radius R , if its radius is now increased to $1.03 R$, find out the percentage change in time period.

Sol. $T^2 \propto R^3$

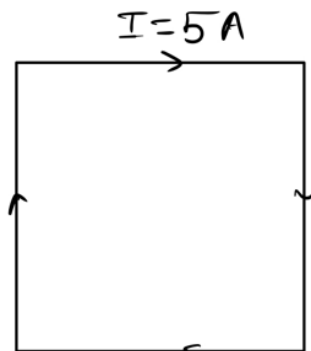
$$\frac{2\Delta T}{T} = \frac{3\Delta R}{R}$$

$$\frac{\Delta T}{T} = \frac{3}{2} \left(\frac{\Delta R}{R} \right) = \frac{3}{2} \times \frac{0.03}{1}$$

$$= 9/2\%$$

$$\Rightarrow 4.5\%$$

- Q7** Find magnetic field at the centre of current carrying square loop of side length $a = \frac{1}{\sqrt{2}} \text{ m}$.



Sol.

$$\frac{\mu_0 i}{4\pi a} (\sin \phi_1 + \sin \phi_2)$$

$$\frac{\mu_0 i}{4\pi \times \frac{1}{2}\sqrt{2}} \times \sqrt{2} \times 4$$

$$= 10^{-7} \times 5 \times 16 = 8 \times 10^{-6} \mu\text{T}.$$

Q8 If a water drop of radius R split into 27 equal parts then work done is 10J. Find the work done when a drop is split into 64 equal parts.

Sol. $\frac{4\pi}{3} \times R^3 = 27 \times \frac{4\pi}{3} \times r^3$

$$R = 3r$$

$$W = S [\Delta A]$$

$$10\text{J} = S[27 \times 4\pi r^2 - 4\pi R^2] \quad \dots(1)$$

$$10 = S \times [4\pi r^2] \left[27 \times \frac{1}{9} - 1\right]$$

$$= 25 \times 4\pi r^2$$

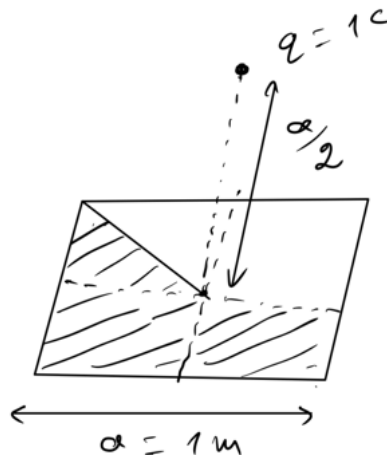
$$\frac{4\pi}{3} R^3 = 64 \times \frac{4\pi}{3} \times r_1^3$$

$$R = 4r_1$$

$$W_1 = S \times [64 \times 4\pi r_1^2 - 4\pi R^2]$$

$$W_1 = S \times \left[64 \times 4\pi \times \left(\frac{R}{4}\right)^2 - 4\pi R^2 \right]$$

Q9 If flux through shaded portion of square is $\frac{P}{48\epsilon_0}$ then find P .



Sol. $\frac{q}{6\epsilon_0} \times \left(\frac{1}{18} + \frac{1}{4} + \frac{1}{4} \right)$

$$P = 5q$$

$$P = 5 \times 1$$

$$= 5$$

Q10 What is relative shift of focal length of a lens with optical power is increased from 0.1 D to 2.5 D?

(1) 24/25

(2) 13/10

(3) 21/25

(4) 11/10

Sol. $f_1 = \frac{1}{0.1} = 10\text{m}$

$$f_2 = \frac{1}{2.5} = \frac{10}{25} \text{ m}$$

$$\frac{|f_2 - f_1|}{f_1} = \frac{10 - \frac{10}{25}}{10} = \frac{24}{25}$$

Q11 In YDSE experiment, two light of wavelength $\lambda_1 = 480 \text{ nm}$ and $\lambda_2 = 600 \text{ nm}$ interfere on the screen which is placed at $D = 1 \text{ m}$ from slit and separation between slits is $d = 1 \text{ mm}$. Find the distance from central maxima where maxima due to both the light first time coincides.

Sol. $y = \frac{n_1 \lambda_1 D}{d} = \frac{n_2 \lambda_2 D}{d}$

$$n_1 \lambda_1 = n_2 \lambda_2$$

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{60}{48} = \frac{5}{4}$$

$$y = \frac{4 \times 600 \times 10^{-9} \times 1}{10^{-5}} = 24 \times 10^{-4} = 2.4 \times 10^{-3} \text{ m}$$

$$2.4 \text{ mm}$$

Q12 Radius of curvature of plano convex lens is 2 cm and refractive index is 1.5 has focal length f_1 in air and f_2 in a medium of refractive index 1.2 . Calculate f_1/f_2 .

Sol. $n_g = 1.5$

$$n_\ell = 1.2$$

$$F_{\text{minima}} = 0.2 \text{ cm}$$

$$\frac{1}{f_{\text{max}}} = \frac{2}{r_{\text{eff}}}$$

$$0.2 = \frac{R_{\text{eff}}}{2}$$

$$R_{\text{eff}} = 0.4 \text{ cm}$$

$$\frac{1}{f_{\text{min}}} = 2 \left(\frac{n_g - n_e}{R} \right)$$

$$\frac{1}{0.2} \Rightarrow \left(\frac{1.5 - 1.2}{R} \right) R \quad R = \frac{0.6}{5}$$

$$5 = 2 \left(\frac{0.3}{R} \right) \quad R = 0.12 \text{ cm}$$

$$5 = \frac{0.6}{R}$$

Q13 Given force $F = \alpha + \beta x^2$, where $\alpha = 1 \text{ N}$ and the work done by this force during a displacement of 1 m is 5 J . The value of β is:

Sol. $\omega = \int \vec{F} \cdot d\vec{x} = \int_0^x (\alpha + \beta x) dx = \alpha x + \frac{\beta x^2}{2}$

$$\omega = \alpha + \frac{\beta}{2} = 5$$

$$1 + \frac{\beta}{2} = 5 \Rightarrow \beta = 8$$

Q14 Temperature of one mole of monoatomic gas increased by 50°C at constant pressure. Then the ratio of Heat supplied to change in internal energy is:

Sol. Const pressure $\Rightarrow$ Isobaric

$$Q = nC_p\Delta T.$$

$$\Delta U = nC_v\Delta T$$

$$\frac{Q}{\Delta U} = \frac{C_p}{C_v} = r = \frac{5}{3}$$

Q15 Energy stored in capacitor of plate area A and separation between plates d is:

$$(1) \frac{1}{2} \epsilon_0 E^2 Ad$$

$$(2) \frac{1}{4} \epsilon_0 EAd$$

$$(3) \frac{1}{8} \epsilon_0^2 E^2 A^2 d^2$$

$$(4) \frac{1}{2} \epsilon_0^2 EAd$$

Sol.

$$\frac{\text{Energy}}{\text{Volume}} = \frac{1}{2} \epsilon_0 E^2$$

$$\text{Total energy} = \frac{1}{2} \epsilon_0 E^2 Ad$$

Q16 A screw gauge has $LC = 0.01$ mm, Now pitch of screw gauge increases by 75% and number of division decreases by 50%. Find the new list count of screw gauge.

Sol. $LC = \frac{\text{pitch}}{\text{Total num. of division}} = \frac{x}{n} = 0.01 \text{ mm} \quad \dots(i)$

$$\frac{1.75x}{0.5n} = LC$$

$$3.5 \frac{x}{n} = (LC)$$

$$= 3.5 \times 0.01 \text{ mm} = LC$$

$$0.35 \text{ mm} = LC$$

Q17 An electron of mass m enters in a region of uniform electric field $\vec{E} = -E_0 \hat{k}$ at $t = 0$ with an initial velocity $\vec{v} = v_0 \hat{i}$. If the de-Broglie wavelength is λ_0 initially, the de-Broglie wavelength at a time t is:

$$(1) \lambda_0 \sqrt{1 + \frac{m^2 v_0^2}{e^2 E_0^2 t^2}}$$

$$(2) \lambda_0 \sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}$$

$$(3) \frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$$

$$(4) \frac{\lambda_0}{\sqrt{1 + \frac{m^2 v_0^2}{e^2 E_0^2 t^2}}}$$

Sol. Velocity at $t = t$ is

$$\vec{v} = v_0 \hat{i} + \frac{eE_0}{m} t \hat{j}$$

$$\lambda_0 = \frac{h}{mV_0} \quad \dots(1)$$

$$\lambda = \frac{h}{m\sqrt{V_0^2 + \frac{e^2 E_0^2}{m^2} t^2}} \quad \dots(2)$$

From equation (1) & equation (2)

$$\lambda = \frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 V_0^2}}}$$

Q18 In a process pressure of the gas is directly proportional to temperature then choose correct option.

A : Process is isochoric.

B : Work done in process is zero.

C : Internal energy increase with increase in temperature.

(1) A and B are correct

(2) A and C are correct

(3) A, B and C are correct

(4) B and C are correct

Sol.

$$P \propto T$$

$$P = KT.$$

Work done in this process is zero

$$W = \int P dv$$

$$Pv = nRT$$

$n \rightarrow$ no of moles

$R =$ gas constant

$T =$ temperature

$\therefore$ The process is isobaric $dv = 0$ No work done process is isobaric :

$$PV = nRT$$

$$KT.V = nRT \text{ This (s) is correct}$$

$$U = \frac{t}{2} nRT$$

$$U \propto T \quad U \uparrow$$

A, B and C are correct